# Homework 2 (Part II): Cryptanalysis of the ECC Toy Scheme and Scalability to Real Curves

Cryptography and Security Course

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## 1 Attack on the ECC Toy Scheme (Discrete Logarithm Problem)

In the toy model developed previously, the elliptic curve is defined over a finite field of reduced order ($p = 17$) with a base point $G = (1, 5)$ whose subgroup order is extremely limited ($N = 16$, accounting for the 15 affine points plus the point at infinity)[cite: 1].

### 1.1 Design of the Attack Algorithm

Since the public key is computed as $Q = d \cdot G$, an attacker who knows $Q$ and $G$ can compromise the secrecy of the private key $d$ by formulating an **exhaustive brute-force attack** (solving the *Discrete Logarithm Problem* - DLP):

1. The algorithm iteratively computes successive scalar multiples of the base point $i \cdot G$ for $i = 1, 2, \ldots, N - 1$.
2. At each iteration, it compares the coordinates of the computed point with the coordinates of the target public key $Q$.
3. As soon as an index $i$ is found such that $i \cdot G = Q$, the attacker has identified the private key $d = i$ with mathematical certainty.

## 2 Comparison Between the ECC Attack and the Statistical Attack on Caesar Cipher

Although both attacks aim to recover a secret key, they present deep structural differences:

- **Mathematical vs. Linguistic Nature:** The attack on the ECC toy scheme exploits the algebraic weakness of a small and finite key space, finding a *deterministic and exact* solution via exhaustive search over the cyclic group. Conversely, the attack on the Caesar cipher relies on the *statistical and probabilistic* properties of natural language (character frequencies via the Chi-squared test).
- **Robustness of the Verification Test:** In the case of ECC, the condition $i \cdot G == Q$ yields an unequivocal binary outcome (true/false). In the Caesar cipher, the quality of the decrypted text must be estimated using statistical approximation metrics, making the attack vulnerable to false positives when dealing with very short ciphertexts.

## 3 Why the Approach is Infeasible on Real Curves (Standard secp256k1)

In real-world applications, such as the standard *secp256k1* curve used in cryptography and cryptocurrencies (e.g., Bitcoin), the parameters are not modulo 17, but defined over a 256-bit prime field:

$$p \approx 2^{256} \approx 1.15 \times 10^{77}$$

The order of the group $N$ is approximately $2^{256}$. The implications of this astronomical scale are decisive:

1. **Infeasibility of Brute Force:** An exhaustive search attack would require, on average, $2^{255}$ point addition operations on the curve. Even assuming the deployment of the world's most powerful supercomputers capable of performing billions of operations per second, the required time would exceed the estimated age of the universe by orders of magnitude.

2. **Ineffectiveness of Generic Algorithms:** Even when resorting to advanced algorithms for the discrete logarithm problem based on the so-called *birthday paradox* (such as *Pollard's Rho* algorithm), the computational complexity only drops to $O(\sqrt{N}) = O(2^{128})$ operations—a limit that remains computationally intractable with current and foreseeable technology.

Consequently, the security of elliptic curve cryptography does not rely on the secrecy of the algorithm or the geometric structure, but on the mathematical intractability of the Discrete Logarithm Problem when the key space is sufficiently large.